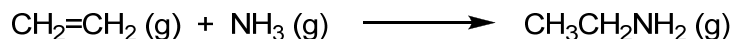


Section B

Answer **two** questions from this section on separate answer paper.

- 5 Ethylamine, $\text{CH}_3\text{CH}_2\text{NH}_2$, is a colourless gas with a strong odour. It can be synthesised by reacting ethene with ammonia, as shown in the equation below:



- (a) (i) State the hybridisation of the C atoms in ethene and in ethylamine.
 (ii) The C-H bonds in ethene are shorter than the C-H bonds in ethylamine. With reference to the hybridisation theory, explain the difference in bond length. **[4]**
- (b) Using relevant data from the *Data Booklet*, calculate a value for the enthalpy change for above reaction. **[3]**

- (c) An experiment was carried out to study the kinetics of the synthesis of ethylamine using ethene and ammonia. The rate equation was found to be

$$\text{rate} = k[\text{CH}_2=\text{CH}_2][\text{NH}_3].$$

When $\text{CH}_2=\text{CH}_2$ and NH_3 , both at an initial concentration of $1.00 \times 10^{-4} \text{ mol dm}^{-3}$, were mixed together at 20°C , $[\text{CH}_3\text{CH}_2\text{NH}_2]$ varies with time, as shown in the table below.

Time / min	$[\text{CH}_3\text{CH}_2\text{NH}_2] / \text{mol dm}^{-3}$	$[\text{NH}_3] / \text{mol dm}^{-3}$	$[\text{NH}_3]^2 / \text{mol}^2 \text{ dm}^{-6}$	$[\text{CH}_2=\text{CH}_2] / \text{mol dm}^{-3}$	Rate / $\text{mol dm}^{-3} \text{ min}^{-1}$
0	0	1.00×10^{-4}	1.00×10^{-8}	1.00×10^{-4}	0.00
5	5.00×10^{-5}				
10	6.90×10^{-5}				
15	7.70×10^{-5}				
20	8.20×10^{-5}				

- (i) Explain the meaning of the term *rate equation*.
 (ii) The rate at a particular time can be approximated using

$$\text{rate} = \frac{\Delta[\text{CH}_3\text{CH}_2\text{NH}_2]}{\Delta t}.$$

Reproduce and complete the above table on your writing paper and hence plot a suitable **straight line graph** to confirm that the reaction shows overall second order kinetics.

[Turn Over]

5 (c) (iii) Hence, determine the rate constant k , with units. [8]

(d) The yield of propylamine from the reaction between propane and ammonia is low. Propylamine can be produced from ethene via another process. Suggest how you can make propylamine from ethene in 3 steps, showing the reagents and conditions necessary as well as all intermediates. [5]

[Total: 20]

[Turn Over]

- 6 (a)** A solid sample contains a mixture of sodium oxide and barium oxide only. The percentage by mass of barium oxide in the sample was determined using a titrimetric method.

The sample of unknown mass was reacted with 50 cm³ of 0.0100 mol dm⁻³ of dilute sulfuric acid, forming a white precipitate, barium sulfate, and a colourless solution. The resultant mixture was filtered. Upon drying, the residue weighed 0.0352 g.

The filtrate was titrated against 0.0200 mol dm⁻³ solution of NaOH. 17.30 cm³ of NaOH was required to completely neutralise the excess sulfuric acid.

- (i)** Write two separate equations, with state symbols, for the reactions between the two oxides and sulfuric acid.
- (ii)** Calculate the mass of barium oxide present in the sample.
- (iii)** Calculate the amount of sulfuric acid that reacted with the sample of solid oxides.
- (iv)** Hence calculate the amount of sulfuric acid reacted with solid sodium oxide only.
- (v)** Calculate the total mass of the sample of solid oxides and hence the percentage by mass of barium oxide in the sample.

[9]

- (b)** Solution **P** was made by dissolving 0.500 g of Na₂O and 0.232 g of BaO in 500 cm³ of water. Solution **P** was then used to determine the concentration of a solution of ethanoic acid via titration.

- (i)** Calculate the pH of solution **P**.
- (ii)** Suggest a suitable indicator for the titration.

[4]

- (c)** Alcohol **F**, C₆H₁₄O, changes orange acidified potassium dichromate (VI) to green, and gives a yellow precipitate with warm aqueous alkaline iodine. **F** reacts with concentrated sulfuric acid at 180°C to give **G** as the only product.

On treating **G** with an excess of hot acidified potassium manganate (VII), compound **H**, C₅H₁₀O₂, is formed.

H gives effervescence with solid sodium hydrogencarbonate.

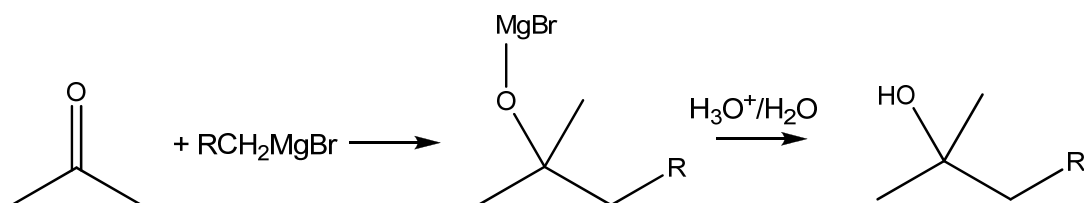
Use the information above to deduce the structural formulae for **F**, **G** and **H**, explaining your reasoning.

[7]

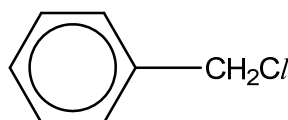
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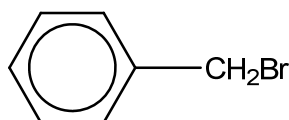
- 7 (a) Grignard reagents are typically used as a reagent to form alcohols from carbonyl compounds.



- (i) Suggest the starting carbonyl compound needed for the formation of pentan-3-ol using $\text{CH}_3\text{CH}_2\text{MgBr}$ as the Grignard reagent.
- (ii) One of the problems associated with the use of magnesium ribbon to produce Grignard reagents is the slow initial step. This is especially so when the magnesium ribbon exposed to air is used. Suggest a reason.
- (iii) A few drops of Universal Indicator turns yellow in an aqueous solution of MgBr_2 . Explain with the aid of relevant equations. [4]
- (b) Benzyl halides are chemical warfare agents used as tear agents. Two such benzyl halides are benzyl chloride and benzyl bromide.



Benzyl chloride



Benzyl bromide

The boiling points of benzyl chloride and benzyl bromide are given below.

Compound	Boiling point ($^{\circ}\text{C}$)
benzyl chloride	179
benzyl bromide	201

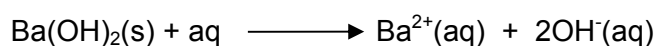
- (i) State the reagent and conditions needed to convert methylbenzene into benzyl chloride. State the type of reaction involved.
- (ii) Suggest a simple test-tube reaction you could use to distinguish benzyl chloride from its isomer, 2-chloromethylbenzene. You should state the reagent used and the observation you would expect to make.

[Turn Over

- 7 (b) (iii) Account for the difference in the boiling points between benzyl chloride and benzyl bromide.
- (iv) State the reagents and conditions needed to convert benzyl chloride to benzaldehyde, $\text{C}_6\text{H}_5\text{CHO}$, in two steps, drawing any intermediate product formed.

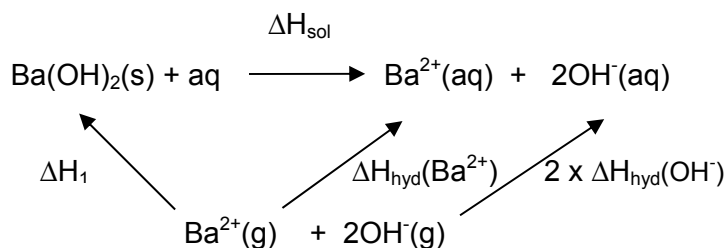
[9]

- (c) The enthalpy change of solution of barium hydroxide, ΔH_{sol} , is defined according to the following equation:



An energy cycle is constructed using the information in the table below.

Enthalpy change	kJ mol^{-1}
Enthalpy change of hydration of Ba^{2+} , $\Delta H_{\text{hyd}}(\text{Ba}^{2+})$	-1505
Enthalpy change of hydration of OH^- , $\Delta H_{\text{hyd}}(\text{OH}^-)$	-664
ΔH_1	-2850



- (i) Identify ΔH_1 .
- (ii) Using the information given and the energy cycle above, calculate the enthalpy change of solution for Ba(OH)_2 .

[3]

[Turn Over]

- 7 (d) The standard enthalpy change of neutralisation was determined experimentally by mixing known volumes of aqueous benzoic acid and aqueous barium hydroxide. The process was known to be only 80% efficient. The following results were obtained.

- Initial temperature = 25.0 °C
- Final temperature = 29.8 °C
- Volume of 1.50 mol dm⁻³ C₆H₅COOH (aq) used = 40 cm³
- Volume of 0.50 mol dm⁻³ Ba(OH)₂ (aq) used = 40 cm³

- (i) Use the above data to calculate the standard enthalpy change of neutralisation of benzoic acid. Assume that the specific heat capacity of all solutions = 4.2 J K⁻¹ cm⁻³
- (ii) Predict, with a reason, if the enthalpy change of neutralisation reaction will be more or less exothermic when hydrochloric acid is used instead. [4]

[Total: 20]

--The End--